

University of Pretoria

MIT 805 Group 19

Group Project Part 2

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I. INTRODUCTION

The TLC Trip Record Dataset contains data collected by the NYC Taxi and Limousine Commission (TLC) on taxi trip records, including information on trip distances, trip fares, and pickup and drop-off locations. This dataset was chosen to be used for large-scale data processing and analysis, through the use of PySpark.

II. DATASET

Data source: New York City Taxi and Limousine Commission (TLC) Trip Record Data (<https://d37ci6vzurychx.cloudfront.net/trip-data>), used under the NYC.gov Terms of Use, and also mirrored on the AWS Open Data Registry. The dataset does not contain any personally identifiable data. The data is published monthly by NYC TLC.

Table I outlines the raw dataset, and the three data subsets used throughout this study. The working dataset, from part 1, contained 139 months of data, while the subset used in part 2 contained 48 months of data (167,858,646 rows in total). The total size of data processed by PySpark was 4.17 GB, spanning 156,964,574 rows in total. The data processed by PySpark is larger than the subset due to the Parquet files being compressed on the disk compared to the data being decompressed into Spark.

The subset chosen was data between January 2022 and December 2025. This subset was chosen to exclude data collected during the COVID-19 period, which would have distorted the data towards abnormal operating conditions if it were included. The subset also contains data from January 2025, during which there was a congestion-pricing change,

TABLE I
DATASET AND SUBSET SIZES

Tier	Size
Raw catalogue	~31 GB
Working set (Part 1)	12.15 GB
Subset (Part 2)	2.58 GB
Processed by PySpark	4.17 GB

making the data relevant to current policies.

Table II provides a summary of the filtering that was conducted on the raw dataset, in order to derive the data subset that was used in this part of the project. Filtering was done to ensure that any incorrect data entries were excluded from the analysis, and that only trips in the specified time period were included. Each filter was applied independently to the dataset, resulting in overlap between the filtered rows. After applying the filters together to the raw dataset, a total of 10,894,072 rows were removed.

One of the challenges with the dataset was in handling some of the inaccuracies present, which resulted in the need to filter the raw dataset. For example, it was found that 0.36% of trips had a recorded drop-off at, or before, the pick-up, resulting in zero or negative trip durations.

III. PROCESSING OBJECTIVE AND ANALYTICAL QUESTION

This study aims to answer the following research questions:
RQ1: Where does congestion cost the yellow taxi fleet the most revenue?

RQ2: Which routes experience the most congestion during peak hours?

Both questions were answered by following the same methodology: a free-flow baseline was first determined, from

TABLE II
DATASET FILTERING

Filter	Rows passing	% retained
Initial rows	167,858,646	100.00%
Time-frame (2022–2025)	167,857,980	100.00%
Drop-off > pickup	167,251,228	99.64%
Duration > 0	167,251,228	99.64%
1 min < duration < 3 hr	165,152,624	98.39%
0.1 mi < distance < 100 mi	163,510,043	97.41%
1 mph < speed < 70 mph	163,174,278	97.21%
\$0.01 < fare < \$1000	163,575,252	97.45%
Zone ID between 1 and 263	165,660,093	98.69%
Final subset	156,964,574	93.51%

which the time-loss per trip was calculated. This value was then converted into a monetary value. The same data records were utilised to answer both questions, with only the grouping key differing between the two executions. For RQ1, the data was grouped by (pickup zone, hour, weekday or weekend), and for RQ2 the data was grouped by (pickup zone, drop-off zone, time window).

For the free-flow baseline, off-peak trip data from the same dataset was used to derive the baseline. For each pickup-dropoff pair, the baseline was determined by analysing weekday trips between 01:00 am and 04:59 am by dividing the total distance travelled by the total duration. An assumption was made that trips during this time window will experience the least congestion during the weekdays, and will therefore provide a baseline that peak-traffic trips can be compared to. To ensure an accurate baseline was generated, the study only included routes with more than 100 trips in the baseline. This resulted in 3,599 routes being included in the study. Each route was compared with its own respective baseline, instead of using an overall average, to ensure that results were an accurate representation of each route. If an average baseline was used throughout, the results may have been skewed by high-traffic routes, such as airport routes, which would still experience a higher volume of trips during the off-peak hours compared to routes in quieter residential areas.

Four measures were obtained after the baselines were constructed: 1. Travel Time Index (TTI): free-flow baseline traffic speed divided by the actual speed. 2. Delay hours: actual trip duration minus the duration of the trip if taken at the free-flow speed. 3. Yield: revenue per hour generated. 4. Revenue lost / forgone: the difference between the estimated baseline yield (without congestion) and the actual yield, multiplied by the total hours worked.

The research questions were chosen to outline the overall effect of congestion on revenue generated across the routes travelled. For yellow taxi trips, the meter charges for elapsed time as well as distance. During peak congestion trips, the fare of the individual trip increases, while the number of trips that the driver completes decreases for that time period. The revenue generated per vehicle hour is therefore affected. For this study, revenue was calculated as the sum of the base fare amount and the tip amount, excluding tolls and surcharges.

IV. MAPPING / TRANSFORMATION

Figure 1 shows the RDD formulation that was used for the map function implementation. The aim of the mapping function is to produce the key that is used to answer the research questions.

Each record produced by the map function carries five values, namely distance, duration, revenue, delay hours, and trip count. These are all summed values, and not averages, which allows the reduce function to be able to add the partial

```
1 pairs = rows.map(lambda r: ((r["pu"], r["hour"], r["daytype"]),
2                             (r["revenue"], r["duration_hr"], 1)))
```

Fig. 1. RDD formulation for mapping

TABLE III
COMPARISON OF REDUCEBY AND GROUPBY OPERATIONS

Operation	Shuffle write	Execution time
reduceByKey	0.207 MB	11.1 s
groupByKey	29.112 MB	14.3 s

results in any order without changing the result. The averages were calculated after the reduction step.

V. SHUFFLE / GROUPING

Rows that share the same key were first grouped together before they could be reduced. This was done by assigning each record to a partition according to a hash of its key. All records which shared a key were assigned to a single partition, and each partition was then processed as a single task. This resulted in an uneven workload in the cases where some keys contained more records than others.

The volume of data crossing the network is largely dependent on whether the reduction is combined with the map function before the shuffle. In order to measure the difference between the reduceBy and groupBy operations, both aggregations were computed over a single month of data. The reduceByKey applied the reduction locally within each partition before the exchange, while the groupByKey did not. The results from this comparison are shown in table III. Both operations produced 10,142 rows of results.

VI. REDUCTION / AGGREGATION

The reduce function must return the same result regardless of the order in which the partial results are combined. In the case of Spark, the partial results were combined concurrently across the tasks, and then once again at the reducer, which did not guarantee a fixed order. In order to work around this challenge, the rates calculated were calculated as sum values, and not as averages. For example, in equation 1, the speed was calculated as the sum of the trip distance, d , divided by the total trip duration, t , for the group. Per-trip averages were not calculated, as using this approach would have resulted in shorter trips holding the same weight as longer trips, whereas computing a ratio of the sums takes into account the distance travelled in the given time-frame and returns a weighted value. The trip yield, defined as the total revenue divided by total trip duration, was calculated similarly.

$$\bar{v}_g = \frac{\sum_{i \in g} d_i}{\sum_{i \in g} t_i} \quad (1)$$

The following values were summed during the shuffle reduction: distance, duration, revenue, delay hours, and trip count. The rates were calculated once, after the completion of the reduction step. The reduction process was verified by first implementing using RDD (using the map and reduceByKey operations), and then mirroring the process with a DataFrame version (using groupBy and aggregate operations). The results were compared and were found to match across all 12,270 rows computed.

The reduction process was executed to produce individual result sets for RQ1 and RQ2. For RQ1, a total of 12,173 rows were produced, and for RQ2 a total of 14,514 rows were produced, each with over 500 trips.

VII. SPARK EXECUTION AND DAG

For the Spark execution, the transformations were collected into a directed acyclic graph (DAG), which is only executed once an action requires a result. Therefore, the transformations were not executed in the exact order in which they were written.

Figure 2 shows the plan that was executed to obtain the results for RQ1, and figure 3 shows the execution plan for RQ2. The execution plan was obtained using the code block shown in figure 4.

The first part of the Spark execution involved scanning the Parquet files to keep only the required rows and columns, and then aggregating the data within each partition. Exchange hash partitioning was then performed on the grouping key, after which the data was again aggregated, performing the shuffle operation. BroadcastHashJoin was used to add the zone lookup data, since the lookup table was fairly small compared to the primary dataset.

The two aggregation stages were also written against the RDD API, which used the reduceByKey. It was found that the RDD execution ran approximately 42 times slower than the DataFrame execution. One possible reason for this difference in execution times could be that the DataFrame execution only read the required columns from the Parquet files, while the RDD execution built a Python object for every row.

Hadoop and Spark differ in the following manner: when working with Hadoop MapReduce, the data output by the map stage is written to the disk, after which the shuffle and reduce stages are conducted as separate jobs. In Spark, results are kept in memory instead of being written to the disk after each stage.

VIII. RESULTS AND TECHNICAL INTERPRETATION

The purpose of this analysis was to identify which NYC taxi routes experience the highest levels of congestion during peak hours and to determine how congestion affects revenue generation. Congestion was measured using the Travel Time

```
1 == Physical Plan ==
2 InMemoryTableScan (1)
3 +- InMemoryRelation (2)
4   +- Project (46)
5     +- BroadcastHashJoin Inner BuildRight (44)
6       :- Project (39)
7         +- HashAggregate (38)
8         +- Exchange (37)
9         +- HashAggregate (36)
10        +- Project (35)
11          +- BroadcastHashJoin LeftOuter BuildRight (34)
12            :- Project (21)
13              +- BroadcastHashJoin LeftOuter BuildRight (20)
14              :- Project (6)
15                +- Filter (5)
16                +- ColumnToRow (4)
17                +- Scan parquet (3)
18              +- BroadcastExchange (19)
19              +- Filter (18)
20            +- InMemoryTableScan (7) (columnIn=false, columnOut=true)
21              +- InMemoryRelation (8)
22                +- Project (17)
23                +- Filter (16)
24                +- HashAggregate (15)
25                +- Exchange (14)
26                +- HashAggregate (13)
27                +- Project (12)
```

Fig. 2. Spark execution for RQ1

```
1 == Physical Plan ==
2 InMemoryTableScan (1)
3 +- InMemoryRelation (2)
4   +- Project (27)
5     +- Project (26)
6       +- BroadcastHashJoin Inner BuildRight (25)
7         :- Project (11)
8         +- Filter (10)
9         +- HashAggregate (9)
10        +- Exchange (8)
11        +- HashAggregate (7)
12        +- Project (6)
13          +- Filter (5)
14          +- ColumnToRow (4)
15          +- Scan parquet (3)
16        +- BroadcastExchange (24)
17          +- Filter (23)
18          +- InMemoryTableScan (12) (columnIn=false, columnOut=true)
19            +- InMemoryRelation (13)
20              +- Project (22)
21                +- Filter (21)
22                +- HashAggregate (20)
23                +- Exchange (19)
24                +- HashAggregate (18)
25                +- Project (17)
26                +- Filter (16)
27                +- ColumnToRow (15)
28                +- Scan parquet (14)
```

Fig. 3. Spark execution for RQ2

Index (TTI), it compares the travel time during peak hours to the travel time under off-peak hours. A higher TTI indicates greater congestion.

Figure 5 shows that the highest levels of congestion occurred on routes connected to business centres and airports. These routes experienced longer travel times during peak periods compared to off-peak hours. It is evident from the results that congestion is concentrated in areas with high

```
1 def capture_plan(df, name):
2     buf = io.StringIO()
3     with contextlib.redirect_stdout(buf):
4         df.explain("formatted")
5     text = buf.getvalue()
6     (LOCAL_OUT / f"plan_{name}.txt").write_text(text)
7     print(text[:1800])
8     return text
9
10 _ = capture_plan(rq1, "rq1_zone_hour")
11 _ = capture_plan(rq2, "rq2_route_window")
```

Fig. 4. Spark execution plan

TABLE IV
STAGE METRICS, SIGNIFICANT STAGES ONLY

Stage	Role	Tasks	Med. (ms)	Shuffle W	Skew
4	Filtering write	29	11,018	7,766 MB	1.3
5	Filtering read	32	16,250	—	3.7
21	RQ2 aggregation	48	1,165	108.3 MB	3.0
37	RQ1 aggregation	48	1,356	17.7 MB	2.5
54	DataFrame twin	48	1,127	11.6 MB	1.3
56	RDD twin	48	52,202	10.2 MB	1.2

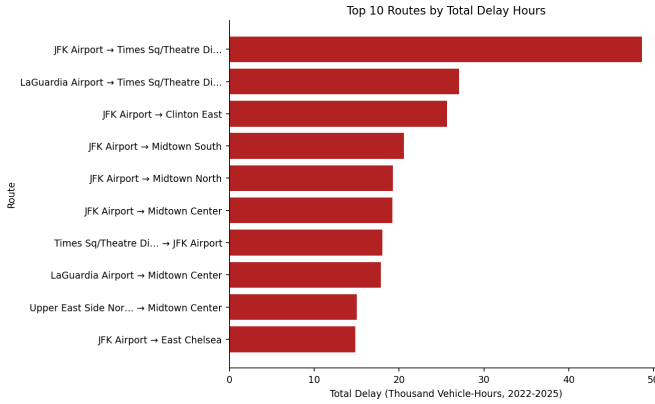


Fig. 5. Top ten most congested routes

travel demand. Business hubs such as Midtown Manhattan and transport hubs such as JFK and LaGuardia airports attract high volumes of commuters and travellers, resulting in increased traffic volumes and travel delays.

It can be seen from figure 6 that congestion was generally higher during the PM peak period than during the AM peak period. Evening travel demand appears to create greater pressure on the road network, mainly around business districts and airports. Drivers are, therefore, more likely to experience delays during afternoon and evening shifts.

The graphs in figure 7 that how congestion affects taxi

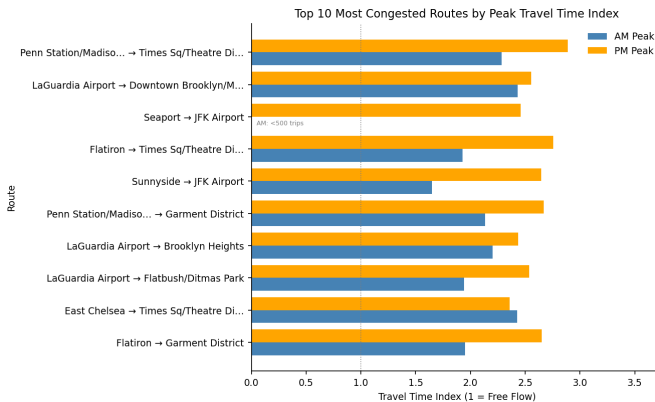


Fig. 6. AM vs PM traffic congestion comparison

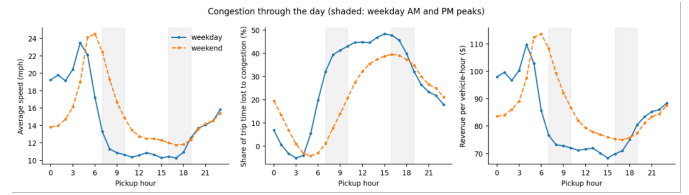


Fig. 7. Congestion through the day

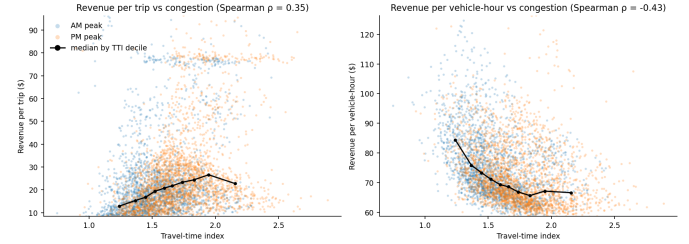


Fig. 8. Congestion vs revenue scatter plot

operations in New York City. During weekday afternoon periods, almost half of total travel time is lost due to congestion, causing average speeds to drop to around 10 mph. These highly congested periods correspond with the lowest revenue per hour, showing that congestion reduces taxi productivity and earning potential. In contrast, low-congestion periods during the early morning hours allow faster travel speeds and generate the highest revenue levels. Managing congestion could improve both transportation efficiency and taxi industry revenue.

The graphs in figure 8 show that congestion has a mixed effect on revenue. It was observed that Revenue per trip increases as congestion increases as passengers spend more time in traffic and fares continue to accumulate.

Revenue per hour on the other hand tends to decrease as congestion becomes more severe. Although drivers earn more from individual trips, they complete fewer trips within the same amount of time. These result show that higher fares do not necessarily lead to higher overall earnings. This suggests that moderate congestion may benefit drivers through higher fares, but severe congestion reduces overall profitability.

To understand the relationship between congestion and earnings better, revenue was analysed across different congestion levels for the same route. From figure 9, revenue per trip increases as congestion increases. When congestion more than doubled compared with the least congested period of the route, revenue per trip increased by roughly 21%. This suggests that longer travel times result in slightly higher fares being charged to passengers. The opposite is true for revenue per hour. Revenue per hour decreases as congestion increase. At the highest congestion levels (TTI=2.0+), revenue per vehicle-hour dropped by roughly 42%. While drivers earn slightly more from individual trips during congested

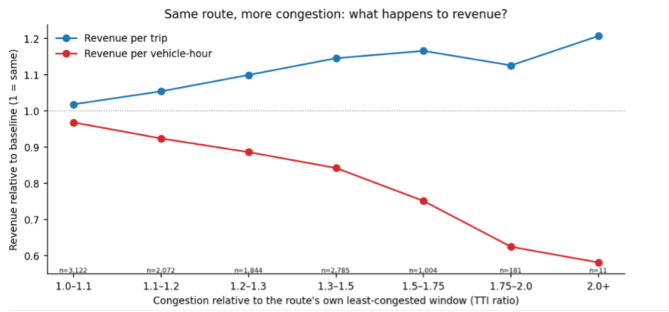


Fig. 9. Revenue versus congestion on the same route

```

-- sensitivity table:
  variant   trips  slowest_hour  min_speed  peak_yield  trough_yield
headline 156964574
no_airports 141989902
pre_2025_only 113501876
2025_only 43462698

```

variant	trips	slowest_hour	min_speed	peak_yield	trough_yield
headline	156964574	17	10.65	105.21	70.24
no_airports	141989902	15	8.46	92.20	67.50
pre_2025_only	113501876	17	10.78	109.87	69.42
2025_only	43462698	15	10.24	99.25	72.35

Airport routes in top 10 by TTI: 5, by delay hours: 9, by revenue: 8
Top-10 by TTI unchanged without airports: 5 of 10 routes
Spearman ρ , TTI vs rev_per_trip, excluding airports: +0.29 (with airports: +0.35)
Spearman ρ , TTI vs rev_per_hr, excluding airports: -0.59 (with airports: -0.43)

Fig. 10. Sensitivity analysis

conditions, the increase in trip revenue is not enough to compensate for the additional time spent in traffic.

A sensitivity analysis, shown in figure 10, was conducted to test whether the findings were robust and not driven by a specific subset of the data, such as airport trips or a particular time period. By conducting the analysis on different datasets (full sample, excluding airports, pre-2025 data, and 2025-only data), the study was able to verify that the relationship between congestion and revenue remained the same. The results are, therefore, reliable.

IX. BUSINESS / SOCIETAL INTERPRETATION AND VALUE

There is a lot of value that can be extracted by different stakeholders from these insights such as Drivers, Taxi companies and Traffic management agencies and most importantly to optimise revenue. Drivers can use these insights to identify routes and operating times that offer the best balance between travel time and earnings. They would also be able to operate more efficiently by avoiding highly congested routes and improve hourly income and reduce unproductive time spent in traffic.

Taxi companies can use congestion information when planning shifts and allocating vehicles. Better route selection can optimise fleet utilisation and reduce operating inefficiencies caused by traffic delays. Traffic management agencies can use the results to identify routes that contribute most to congestion. These routes can be prioritised for traffic management measures, infrastructure improvements and congestion-reduction strategies.

X. LIMITATIONS

Several limitations should be considered when interpreting the findings: Speed was calculated using trip distance and trip

duration rather than actual GPS traffic measurements. Cash tips were not recorded in the dataset and therefore actual driver earnings may be higher than reported. Weather conditions, accidents, road closures and special events were not included in the analysis. The study focused only on NYC Yellow Taxi data and excluded ride-hailing services such as Uber and Lyft. The analysis identifies relationships between congestion and revenue but does not prove cause-and-effect relationships.

XI. CONCLUSIONS

This study found that the highest levels of congestion occurred on NYC taxi routes connected to major business and airport areas, with evening peak hours experiencing more congestion than morning peak hours. Although congestion can increase earnings from individual trips, heavy congestion reduces revenue per hour because drivers spend more time in traffic and complete fewer trips. In conclusion, moderate congestion may improve trip revenue, but severe congestion lowers productivity and profitability.

XII. GITHUB REPOSITORY AND VIDEO PRESENTATION

The project setup and code can be found in the following GitHub repo: <https://github.com/ShreyaB143/MIT805-GroupProject>

The recorded presentation can be found through the following link: https://drive.google.com/file/d/1RHd2-ZUbhIqOVprFXTNiTotugnPBbPRI/view?usp=share_link